RetinaGuard AI

Explainable AI for Diabetic Retinopathy Screening in Rural India

A screening-support prototype that checks retinal image quality, estimates DR severity, explains the prediction visually, and supports referral to an ophthalmologist.



Problem & track

Team ID: TS -0123

• Track: MedTech

Team details

• Team: TecnoX

• ID: TS -0123

TEAM MEMBERS

#	Full name	Roll number	Branch & year	Role
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Disclosure

• AI tools used for template layout and language adaptation.

• Confirm disclosure with the team.

• Sources listed on slide 8.

Problem depth

Who is affected

• Adults with diabetes in India, especially those screened at PHCs and outreach camps.
• People capturing fundus images who need reliable quality feedback.
• Ophthalmologists with limited review capacity.

How bad it is

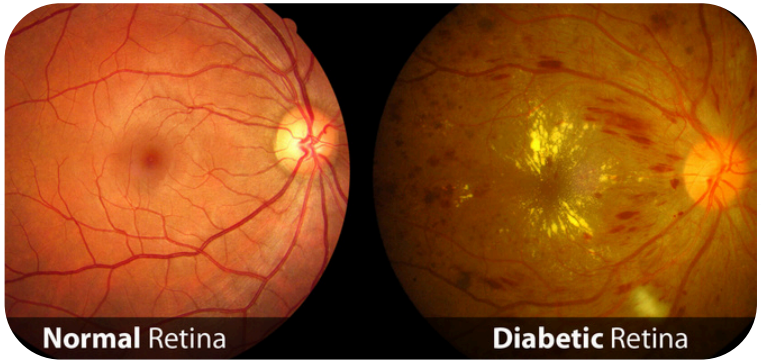
• DR may progress before obvious symptoms and can cause permanent vision loss.
• Limited specialist access can delay screening and assessment.
• Portable-camera image quality can vary in field conditions.

Evidence

• Supplied PS 26038 brief reports 77M+ adults with diabetes in India and estimates DR affects about 18%.
• Source: organiser-provided problem statement.
• VERIFY figures against primary sources before final submission.

What happens if nothing changes

• Patients needing professional assessment may be identified late.
• Frontline teams may lack consistent quality feedback and an interpretable report.
• Delays can increase risk of preventable vision loss; this tool supports screening, not final diagnosis.



Existing solutions and the gap

#	Existing solution / source	What it does	Context / limitation to verify	How ours differs
1	LumineticsCore Digital Diagnostics	AI-based DR screening. digitaldiagnostics.com	Check official regulatory scope operating conditions and intended use.	Proposed workflow emphasizes quality feedback , Grad-CAM and referral support.
2	EyeArt EyeNuk	AI screening from retinal images. eyenuk.com	Verify official product scope and evidence; don't assume shortcomings.	Combines quality gating, severity estimate, explanation and report in one prototype.
3	APTOS 2019 dataset	Labeled fundus images for model development and benchmarking.	Dataset, not a clinical product; confirm licensing and split design.	Add image-quality assessment and human-in-the-loop reporting.
4	IDRiD / DRIVE	IDRiD: grading/lesion labels; DRIVE: vessel segmentation.	Different datasets serve different tasks; neither alone validates full pipeline.	Use only where labels and terms match the evaluation task.

The gap we fill

- A rural-screening workflow combining quality assessment, severity estimation, Grad-CAM visualization and referral support—withan ophthalmologist in the loop.

Solution and technical approach

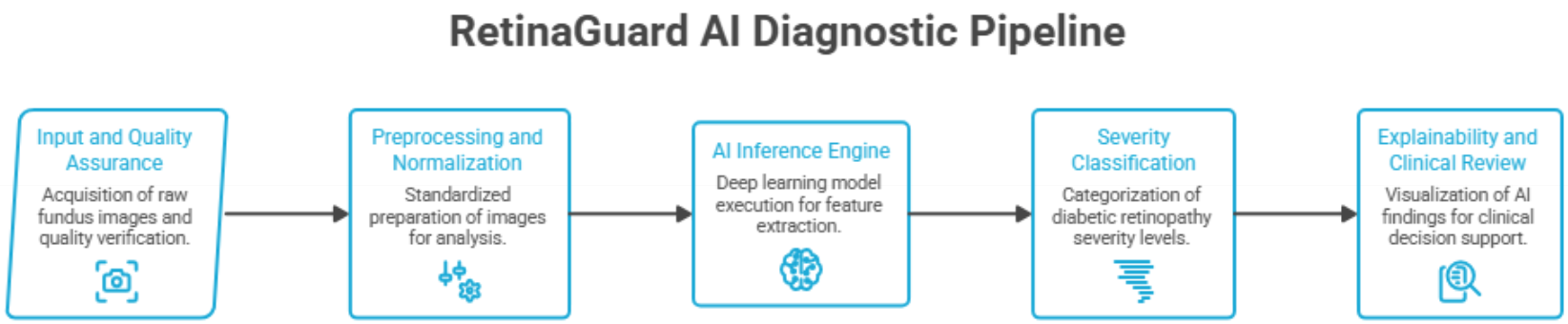
1. **Capture / upload** — Worker uploads fundus photo
2. **Quality check** — Focus • illumination • field of view
3. **Preprocess** — Enhance usable images; reject ungradable
4. **EfficientNet-B0** — Severity estimate + class probabilities
5. **Grad-CAM** — Highlight regions influencing prediction
6. **Referral report** — Ophthalmologist reviews and decides

Technology choices and why

- MATLAB + Image Processing Toolbox: quality checks, CLAHE, illumination normalization and suitable denoising.
- Deep Learning Toolbox: transfer learning with EfficientNet-B0.
- Grad-CAM: visual explanation of regions influencing model output.
- Statistics and Machine Learning Toolbox: sensitivity, specificity, F1, ROC-AUC and confusion matrix.

What is new here

- AI-based DR screening already exists; our proposed contribution is the workflow integration.
- Quality gate → severity estimate → visual explanation → referral-oriented report.
- Planned extensions: optic-disc/fovea localization, vessel segmentation and lesion-level a nalys is.
- Do not claim these extensions are implemented unless demonstrated.



1 System Architecture

Users

Healthcare Worker (Upload & Screen)

Ophthalmologist (Review & Validate)

Frontend

React.js + Tailwind CSS

Responsive UI Dashboard • Upload Reports

Backend API

FastAPI

- Authentication
- Image Processing
- Model Inference
- Report Generation
- Database Management

Image Processing

OpenCV + Pillow

- Quality Check
- Preprocessing
- Normalization
- Tensor Conversion

AI Model

PyTorch
EfficientNet-B0 + Grad-CAM

- DR Classification
- Explainability

Database

PostgreSQL

- Patient Records
- Screening History
- Reports
- Secure Storage

Deployment: Docker (Cloud / Local)

2 End-to-End Workflow

1. **Login** — Healthcare worker logs in and accesses dashboard.
2. **Patient Registration** — Enter patient details (only necessary information).
3. **Image Upload** — JPG / PNG; Preview; Validation
4. **Image Quality Check** — Blur detection; Contrast/illumination; Field of view check
   - If quality is poor: Request recapture
5. **Preprocessing** — Resize; Normalize; Tensor conversion
6. **AI Classification** — EfficientNet-B0; Softmax Probabilities; Severity Category
7. **Grad-CAM** — Highlight key areas; Visual explanation
8. **Screening Report** — Prediction + Probability; Explanation (Grad-CAM); Referral Recommendation
9. **Ophthalmologist Review** — Verify image & result; Clinical decision
10. **Secure Storage** — Save records; Access control; Audit logs

3 Tech Stack

Frontend	React.js + Tailwind CSS
Backend	FastAPI (Python)
Image Processing	OpenCV + Pillow
AI / ML	PyTorch + EfficientNet-B0 + Grad-CAM
Database	PostgreSQL

4 Database Schema (Simplified)

Patients
patient_id (PK)
name
age
diabetes_history
created_at

1 — N

Screenings
screening_id (PK)
patient_id (FK)
image_path
prediction
probability
recommendation
screening_date

1 — N

Users
user_id (PK)
name
role
email

1 — N

Reports
report_id (PK)
screening_id (FK)
summary

5 Key Features

- **Image-quality gating** — Prevents inaccurate predictions from poor images.
- **Explainable AI** — Grad-CAM visualization for transparency.
- **Rural screening workflow** — Designed for trained frontline workers.
- **Referral support** — Helps identify cases needing professional evaluation.
- **Secure & role-based access** — For healthcare workers and ophthalmologists

Security and privacy



Data we handle

• Fundus image; image-quality metrics; severity and class probabilities; Grad-CAM map; generated report.
• Use a coded case ID only if needed. Avoid names, phone numbers and Aadhaar by default. • Purpose: screening, explanation and professional review.

Storage and retention

• Prefer temporary processing in the prototype; store only if required for an approved pilot. • If stored: encryption, role-based access and a defined retention policy.
• Retention period and storage location must follow institutional policy and approvals.

Threat	Protection
Unauthorized access	Role-based access; encryption at rest; strong authentication.
Interception	HTTPS/TLS; no public links for clinical images.
Metadata / re-identification	Remove unnecessary metadata; use coded IDs.
Wrong output / overreliance	Show limitations; require clinician review and follow-up plan.
Evaluation data leakage	Patient-level train/test separation; restrict dataset access.
Malicious upload	Validate file type/size; safely process uploads.

Feasibility, cost and business model

Build effort

- Skills: MATLAB/image processing, ML, reporting/UI and evaluation.
- Student prototype timeline: 2–3 days (ASSUMPTION; depends on scope and access).
- Stage it: quality gate → model → Grad-CAM/report → evaluation.

Running cost

- Not yet measured.
- ASSUMPTION: depends on MATLAB/toolbox licences, camera access, computing hardware and storage.
- Use existing university licences/lab hardware if available.

Dependencies

- MATLAB and required toolboxes.
- Public/de-identified labelled images; verify dataset terms. • Camera access and expert labels for evaluation.
- Ethics/privacy approval before patient data.

Business model

- Potential users: PHCs, screening camps, NGOs, eye-care networks and public-health programmes.
- Possible sustainability: institutional licensing/service contracts only after validation and required a ppro va ls .
- Do not market as autonomous diagnosis.

Smallest version we can build first

- Upload → quality check → supported severity classes → Grad-CAM → report.
- Evaluate on an unseen held-out dataset and report metrics honestly.
- Defer patient deployment, automated lesion segmentation and district-scale claims until validated.

Impact with measurable outcomes

MARKS: IMPACT —10

Outcome	Today (baseline)	Target	How we will measure it
Image-quality feedback	Not measured	Feedback for every uploaded image	Test on labelled gradable/ungradable images; track false accept/reject.
Report generation time	Not measured	≤30 sec per usable image (ASSUMPTION)	Time 100+ test runs on documented hardware; report median and 95th percentile.
Referable-DR performance	Not measured	Aspirational: >90% sensitivity and >85% specificity; not ac hie ved /c la imed	Unseen expert-labelled test set; sensitivity, specificity, precision, F1, ROC-AUC and confusion matrix.
Explanation usefulness	Not measured	Clinician-rated usefulness; target to define	Predefined rubric and ophthalmologist feedback.

Who benefits

- Rural adults with diabetes, frontline healthcare workers and ophthalmologists.

Reach

- Pilot reach is not known. Any planned count is an ASSUMPTION; first validate on held-out data.

Research and references

#	Source / title	Link	Claim / use
1	Problem Statement ID 26038 —MathWorks / organiser brief	Organiser-provided document	Problem framing and supplied statistics; verify primary references.
2	LumineticsCore —Digital Diagnostics	https://www.digitaldiagnostics.com/	Existing AI-based DR screening solution.
3	EyeArt —EyeNuk	https://www.eyenuk.com/	Existing AI-based retinal screening solution.
4	APTOS 2019 Blindness Detection	https://www.kaggle.com/c/aptos2019-blindness-detection	Potential labelled fundus dataset; verify terms.
5	IDRiD —Indian DR Image Dataset	https://ieeexplore.ieee.org/document/8335271	Potential grading/lesion annotation resource; verify access.
6	DRIVE —Retinal Vessel Extraction	https://drive.grand-challenge.org/	Vessel segmentation resource; not full DR validation.
7	Me ssid o r- 2	https://www.adcis.net/en/third-party/messidor2/	Potential dataset; verify terms and labels.
8	Selvaraju et al. (2017), Grad-CAM, ICCV	https://arxiv.org/abs/1610.02391	Method for visual explanations from deep networks.


thank
you